LESSON 3.2

Understand Rates and Unit Rates

Set B · Version A · Second form · supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can describe a rate using its two different units and use a model to find a travel time, and I can find a unit rate to compare prices and decide the better buy.

WHY THIS MATTERS You're comparing ticket prices at different game booths in the arcade. Booth A charges $3 for 5 games, and Booth B charges $5 for 8 games. Which booth gives you a better deal? To find out, you need to figure out the cost per game — that's the unit rate!

REMEMBER

Understand Rates and Unit Rates

Rates & Unit Rates. Formula: Unit Rate = (Total Quantity) ÷ (Number of Units) A rate compares two amounts with different units. A unit rate tells you the amount for just 1 of something, like the cost for 1 game.

1. Rate: A ratio comparing two quantities with different units (e.g. $12 for 3 lbs).
2. Unit Rate: A rate simplified to a denominator of 1 unit ($4 per 1 lb).
3. Divide numerator by denominator to compute the per-one amount

Words: **Rates and Unit Rates *(Tasas y tasas unitarias)*** · **Rate *(Tasa)*** · **Unit Rate *(Tasa unitaria)*** · **Per *(Por)*** · **Ratio *(Razón)***

PART 1 Understand the idea

models and meaning

1 **Circle the letter of the best answer. Show how you know.**

What does the ratio 3:1 mean in words?

(A) The first amount is 3 more than the second

(B) There are 4 things altogether

(C) For every 3 of the first amount there is 1 of the second

(D) The two amounts are equal

SHOW YOUR WORK

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

A bag holds 4 red marbles and 6 blue marbles. Write the ratio of red marbles to blue marbles in simplest form.

(A) 2:3
(B) 3:2
(C) 4:10
(D) 10

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

A bowl holds 4 apples and 6 oranges. What is the ratio of apples to the TOTAL pieces of fruit?

(A) 4:10
(B) 4:6
(C) 6:10
(D) 10:4

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Which pair of amounts does the ratio 2:5 compare?

(A) Two different quantities
(B) A number and its opposite
(C) A total and its parts added
(D) The same quantity twice

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

The arcade snack bar sells popcorn in two sizes: a small bag with 3 cups for $2.25 and a large bag with 8 cups for $5.60. You want the most popcorn for your money. What is the unit rate for the small bag?

(A) $0.70 per cup
(B) $0.75 per cup
(C) $1.33 per cup
(D) $2.25 per cup

PER 1	

1. **Label** *Write what each column is.*
2. **Find per 1** *Divide to get one.*
3. **Scale** *Multiply up to what is asked.*
4. **Check** *Does the size make sense?*

6 **Circle the letter of the best answer. Show how you know.**

The arcade snack bar sells popcorn in two sizes: a small bag with 3 cups for $2.25 and a large bag with 8 cups for $5.60. You want the most popcorn for your money. What is the unit rate for the large bag?

(A) $0.70 per cup
(B) $0.75 per cup
(C) $1.43 per cup
(D) $5.60 per cup

PER 1	

1. **Label** *Write what each column is.*
2. **Find per 1** *Divide to get one.*
3. **Scale** *Multiply up to what is asked.*
4. **Check** *Does the size make sense?*

EXPLAIN YOUR THINKING

How does finding the unit rate help you compare the booths?

SENTENCE STARTER The best deal is Booth ___ because its unit rate of ___ per game is the ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

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PART 1 Understand the idea

models and meaning

1 **Write the letter of the matching item on each line.**

Match each arcade deal to its unit rate (coins per play).

_____ 80 coins for 4 plays

_____ 63 coins for 7 plays

_____ 48 coins for 6 plays

_____ 55 coins for 5 plays

(A) 20 coins per play

(B) 9 coins per play

(C) 8 coins per play

(D) 11 coins per play

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

The arcade snack bar sells popcorn in two sizes: a small bag with 3 cups for $2.25 and a large bag with 8 cups for $5.60. You want the most popcorn for your money. Which size is the better deal, and why?

(A) The small bag, because $2.25 is less than $5.60

(B) The small bag, because it has fewer cups

(C) The large bag, because $0.70 per cup is less than $0.75 per cup

(D) They are the same deal

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

The Star Blaster cabinet scored 90 points in 3 games. What is the unit rate in points per game?

(A) 3 points per game
(B) 30 points per game
(C) 93 points per game
(D) 270 points per game

PER 1

4 **Circle the letter of the best answer. Show how you know.**

Nova: 78 points in 6 min. Pulse: 75 points in 5 min. Which scores faster (more points per minute)?

(A) Pulse, at 15 points per minute
(B) Nova, at 13 points per minute
(C) They score at the same rate
(D) There is not enough information

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

Machine A: 150 coins for 5 plays. Machine B: 168 coins for 8 plays. Which is the better value (fewer coins per play)?

(A) Machine A, at 30 coins per play
(B) Machine B, at 21 coins per play
(C) They are the same value
(D) Machine A, because it costs fewer coins total

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

6 Find the mistake. Explain it, then show the correct work.

Level 2 Extension — Inverted Unit Rate: Points per Minute

1. **Find the scoring rate in points per minute: 96 points in 6 minutes** 96 points in 6 minutes
2. **Set up the division** 6 ÷ 96 = 0.0625
3. **Write the unit rate** The unit rate is 0.0625 points per minute

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

How does finding the unit rate help you compare the booths?

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